A3-PS019-TC072

NDEMS: National Digital Evidence Management System

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Problem Description

Citizen Privacy Leaks (DPDPA 2023):
Case files and chat logs often contain personal details without automatic detection or redaction.

Risk of Evidence Tampering:
Normal storage like pen drives and local servers cannot prove if a file was edited or altered.

Slow Court Approval (BSA 2023 Sec 63):
Proving evidence authenticity manually in court takes too much time and effort.

No Verifiable Record:
Existing tools scan data but cannot create an unchangeable, permanent proof of evidence history.

Background

The Urgent Need for Tamper-Proof Evidence and Privacy Compliance

Massive Digital Evidence Growth:
Police and investigative agencies process terabytes of digital logs and disk images daily, making manual inspection impossible.

Strict Legal Pressure:
Investigators must redact citizen personal data under DPDPA 2023 while proving strict electronic evidence authenticity under BSA 2023 (Section 63).

Broken Tool Ecosystem:
Current forensics tools only inspect raw files—they completely lack an automated bridge to discover private data and anchor chain-of-custody onto an immutable ledger.

Literature & RFC Review

Evaluation of Relevant Security Standards and Forensics Frameworks

01



RFC 3161
(Time-Stamp Protocol)

Proves when evidence was recorded using digital timestamps, but cannot detect or verify privacy redactions.

02



ISO/IEC 27037
(Digital Evidence Handling)

Sets clear guidelines for preserving forensic data, but offers no automated workflow for DPDPA privacy compliance.

03

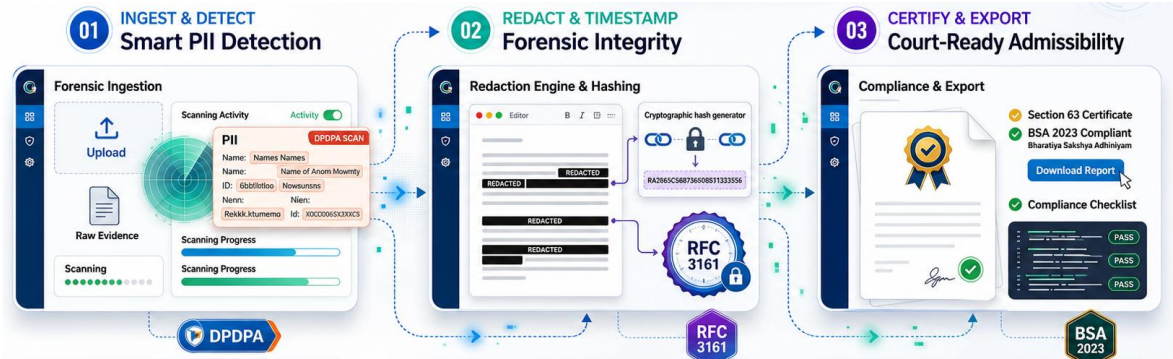


Current PII Scanners
(e.g., Presidio)

Accurately identify personal data in files, but lack tamper-proof audit logs required for court admissibility under BSA 2023 [2].

Proposed Solution

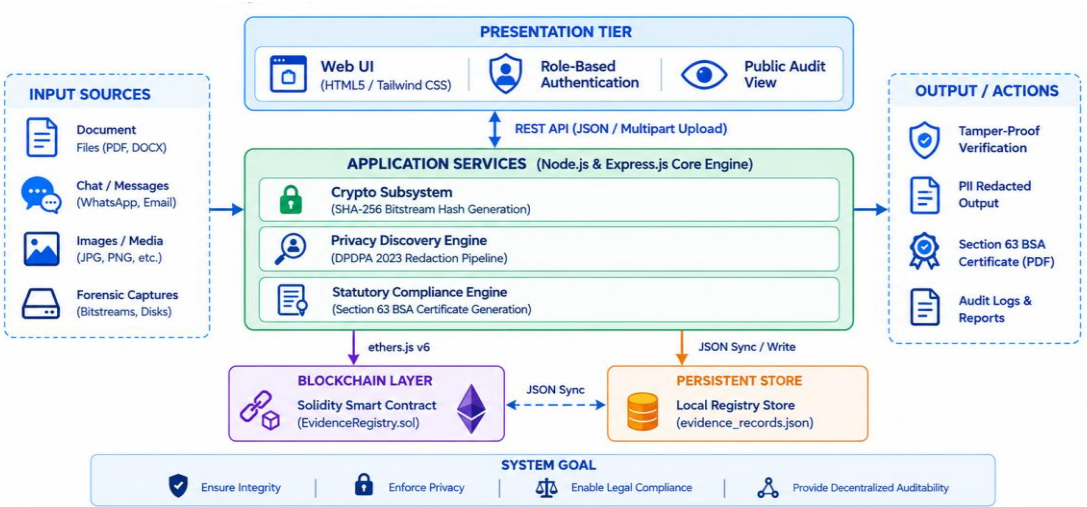
Automated, Privacy-Preserving Digital Forensics for DPDPA & BSA 2023 Compliance



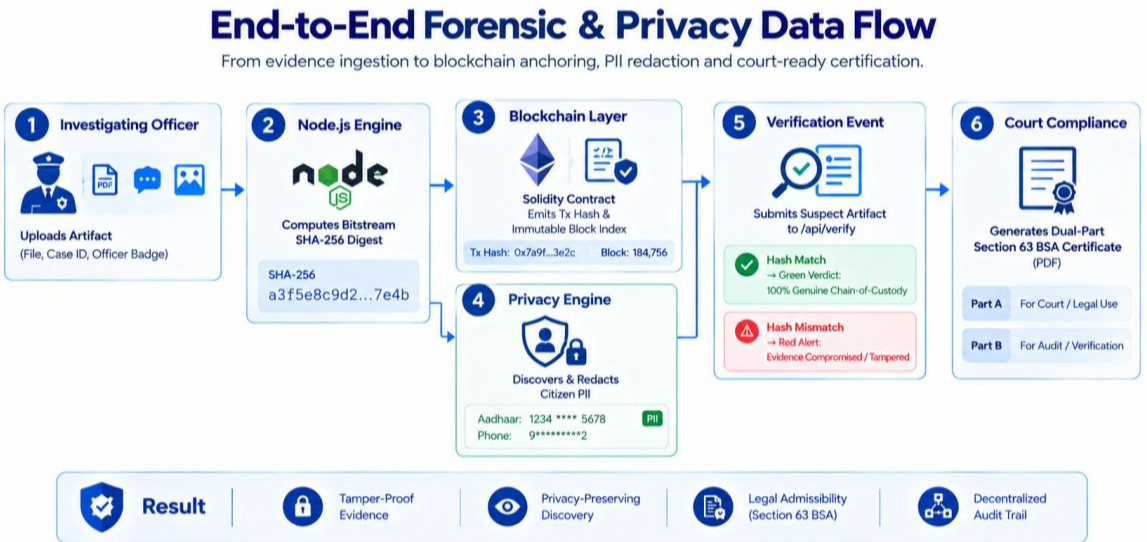
Optimization Proposed

Area	Before (Current)	After (Proposed)	Impact / Benefit
Smart PII Detection	Basic keyword & regex matching, high false positives.	AI/ML based contextual detection with NER & pattern scoring.	Higher accuracy, fewer false positives.
Automated Redaction	Manual redaction (time-consuming, inconsistent).	Rule + AI based auto-redaction with confidence scoring.	Saves time, ensures consistent privacy.
Forensic Integrity	Basic hashing (no time-stamp proof).	RFC 3161 time-stamping + cryptographic hashing.	Stronger chain of custody, court admissible.
Audit & Compliance	Manual logs, scattered reports.	Automated tamper-proof audit logs & compliance reports.	Meets DPDPA & BSA 2023 requirements.
Performance & Scalability	Slow processing, limited scalability.	Optimized pipeline with parallel processing & caching.	Faster processing, handles large evidence volumes.

Solution Architecture & Design



Data Flow Diagram



Deployment Timeline

Phase 1: Core Engine & Testing (Months 1–2)

Set up the offline scanning system, test automated privacy masking (ID numbers, PAN), and process large test drives (500GB+).

Phase 2: Security & Tamper-Proofing (Month 3)

Audit blockchain contracts and integrate official digital timestamps to guarantee files cannot be altered.

Phase 3: Real-World Trials (Months 4–5)

Run pilot trials in forensic labs to generate and verify legal court certificates (Section 63 BSA) on actual case samples.

Phase 4: Full Deployment (Month 6+)

Connect the platform directly to national police and court databases (CCTNS / e-Courts) for seamless nationwide access.

Phase 5: Officer Training (Month 7–8)

Conduct hands-on training workshops for investigating officers and forensic lab staff to streamline daily evidence handling.

Phase 6: Auditing & Maintenance (Month 9+)

Perform quarterly cryptographic audits, update PII detection models for new document types, and ensure ongoing DPDPA compliance.

Bibliography

1. RFC 3161 — Trusted Digital Timestamping (IETF)

Provides tamper-proof cryptographic timestamps to verify the exact creation time of evidence.

2. Microsoft Presidio — Context-Aware PII Masking

Open-source NER framework adapted for automatic detection and redaction of Indian IDs (Aadhaar, PAN).

3. ISO/IEC 27037 — Digital Evidence Handling

International guidelines followed for secure collection, integrity preservation, and chain-of-custody tracking.

4. BSA 2023 (Section 63) — Court Admissibility Framework

Statutory legal standards implemented to generate automated, compliant digital evidence certificates.

Thank You!

Questions & Discussion

NDEMS: National Digital Evidence Management System

Tamper-Proof Digital Forensics with Dual-Certificate BSA 2023 & DPDPA

Team Code – A3-PS019-TC072

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- **Team Member:** Anushka Pandit
- **Faculty Mentor:** Mr. Vishal Bhatt

GitHub:

https://github.com/arunpbt1025/ndems